

Mixed applications



- 1 **a** $h = 86.52$ **b** $a = 20.37$ **c** $a = 10.00$ **d** $y = 2.4$

- 2 $l = 20.6 \text{ m}$

- 3 **a** $x = 149.12^\circ$ **b** $\theta = 102^\circ$ **c** $\theta = 146^\circ$

- 4 **a** $\theta = 37.22^\circ$ **b** $d = 3.4 \text{ cm}$

- 5 $x = 22^\circ$

- 6 **a** $AE = 13.8577$ **b** $BE = 7.6433$ **c** $AB = 21.50$ **d** $BD = 6.3$

- 7 **a** $73\,500 \text{ m}$ **b** $h = 41\,101 \text{ m}$

- 8 $h = 4.4 \text{ m}$

- 9 **a** 74 **b** $d = 28.35$ **c** $h = 16.66$

- 10 **a** $\angle ACB = 102^\circ$ **b** $b = 226.55 \text{ m}$ **c** $h = 117 \text{ m}$

- 11 **a** $x = 23.88 \text{ m}$ **b** $k = 25.95 \text{ m}$ **c** 65.95 m

- 12 **a** 21.89° **b** 123.11° **c** $AB = 29.2$

- 13 **a** 136° **b** $SW = 234 \text{ km}$ **c** $h = 102.58 \text{ km}$

- 14 **a** 23° **b** $a = 35.65$ **c** $h = 20.4$

15

a 3799 km

b 4915.5 km



c $x = 4014.7 \text{ km}$

16 $\theta = 33.56^\circ$

17 $x = 37.79 \text{ km}$

18

a $x = 48^\circ$

b 132°

19

a $\angle ABC = 90^\circ$

b $x = 27^\circ$

c 308°T

20

a $\frac{3}{2}x^2 \text{ cm}^2$

b $\frac{1}{4}$

21

a $\theta = 106.3$

b $x = 36.9$

c $d = 12.0$

22

a $\angle ASC = 90^\circ$

b $AC = 2222 \text{ km}$

c $x = 36^\circ$

d 195°T

23

a $x = 26.73 \text{ km}$

b $b = 20^\circ$

c $\text{N } 34^\circ \text{W}$

24 $x = 41$

25 $y = 2\sqrt{7} \text{ cm}$

26

a 47.13 m

b 76.3 m

c $\theta = 23^\circ$

27

a $\theta = 52.6^\circ$

b 136.66 cm^2